PS200C: Quantitative Methods

In-class Activity 5: Estimation under Selection on Observables

Name: ______________________________

Question 1: All roads lead to the same place

Under CI, every method we discussed is estimating

$$\mathbb{E}_{X\sim F}[\mu_1(X) - \mu_0(X)]\,, \qquad \mu_d(x) \equiv \mathbb{E}[Y(d) \mid X = x].$$

(a) Which distribution F corresponds to the ATE? The ATT? The ATC?

(b) What is the key difference between any estimand of this kind and the simple difference in means? For a given choice of F, how does this differ from what you would get by comparing a group average for the treated vs. control? Why is this what we want to do when we are "adjusting for X"?

Question 2: IPW by hand

Recall that the (fitted) propensity score, $\hat{\pi}(X_i)$ is an estimate of the probability unit i takes the treatment, based on its X_i, i.e. $Pr(D_i = 1 \mid X_i = 1)$. Four units, each with an estimated propensity score $\widehat{\pi}(X_i)$:

Unit	D_i	$\widehat{\pi}(X_i)$	Y_i
1	1	0.80	10
2	1	0.20	6
3	0	0.80	8
4	0	0.20	4

(a) Compute the simple IPW weight $w_i = D_i/\widehat{\pi}(X_i) + (1 - D_i)/(1 - \widehat{\pi}(X_i))$ for each unit.

(b) Which unit gets upweighted the most, and why does that make intuitive sense?

Question 3: Matching the method to the problem

For each scenario, name one estimation approach you'd prefer and one you'd avoid, with a one-line reason. Some choices include sub-classification/stratification, matching on X, matching on the pscore, IPW, balancing weights, and regression.

(a) Just 3 discrete covariates (sex, age bin, party ID), $n = 5000$, with plenty of both treated and control in every cell.

(b) You have 10 covariates and $n = 300$. Treated units are rare in some covariate combinations. However, you suspect (or hope) that not all variables really influence $Pr(D)$ much.

Question 4: Regression as imputation

In the g-computation / regression-imputation view of regression:

(a) What do you fit $\widehat{\mu}_1(x)$ on, and what does it estimate? Same question for $\widehat{\mu}_0(x)$.

(b) When treatment effects vary with X, why is fitting *separate* linear models on treated and control (or equivalently, the Lin regression) preferable to pooled OLS with $Y \sim D + X$?

Question 5: The big picture

Does using a more sophisticated adjustment technique, and/or having a larger sample, help reduce concerns about omitted confounders? Why or why not?